

Multiple Choice (10 marks)

1. A particle moves with $v = \text{const.}$ along the curve $r = k(1 + \cos \theta)$ (a cardioid). Find $\ddot{\mathbf{r}} \cdot \mathbf{e}_r = \mathbf{a} \cdot \mathbf{e}_r$.
 - (a) $-\frac{v^2}{k}$
 - (b) $-\frac{3}{4} \frac{v^2}{k}$
 - (c) $-\frac{7}{4} \frac{v^2}{k}$
 - (d) $\frac{1}{4} \frac{v^2}{k}$
 - (e) $-\frac{1}{4} \frac{v^2}{k}$
2. Find the ratio of the focal lengths of a glass lens in water and in air. The refractive indices of the glass and water are 1.5 and 1.33 respectively.
 - (a) 2.50
 - (b) 0.89
 - (c) 2.23
 - (d) 2.75
 - (e) 4.5
3. Find the dimension of the parallelepiped of maximum volume circumscribed by a sphere of radius R .
 - (a) $\frac{2}{\sqrt{2}}R$
 - (b) $\frac{3}{\sqrt{2}}R$
 - (c) R
 - (d) $\sqrt{2}R$
 - (e) $\frac{1}{\sqrt{3}}R$
4. A uniformly charged conducting sphere of 1.2 m diameter has surface charge density $8.1\mu\text{C}/\text{m}^2$. Find the net charge on the sphere.
 - (a) $97\mu\text{C}$
 - (b) $37\mu\text{C}$
 - (c) $77\mu\text{C}$
 - (d) $87\mu\text{C}$
 - (e) $17\mu\text{C}$

5. An airplane with its nose pointing north with an airspeed of 40 km/h in a 30-km/h crosswind (at right angles) has a groundspeed of
- (a) 40 km/h
 - (b) 60 km/h
 - (c) 50 km/h
 - (d) 10 km/h
 - (e) 90 km/h

True / False (10 marks)

Answer True or False

- 6. True or False: A planet that shrinks to one-tenth its original diameter would cause a person on its surface to weigh 1/10 as much.
- 7. True or False: In a series circuit, adding more lamps will cause the current from the power source to increase due to higher voltage availability.
- 8. True or False: The distance between the object and its image in a plane mirror is 200 cm when the object is 100 cm from the mirror.
- 9. True or False: Satellites are propelled by constant rocket thrust to maintain their orbits around Earth.
- 10. True or False: During an isothermal expansion, 150 J of heat is added to the gas to compensate for the work done on the surroundings.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the factors influencing the intensity ratio of the primary maximum to the first secondary maximum in a diffraction grating with 100 lines, and derive the relevant expressions.
- 12. Calculate the work done in joules when lifting a 5-kilogram mass to a height of 2 meters, and discuss the underlying principles of work and energy involved in this process.

End of Paper